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Experiment 06

Design and Performance Analysis of a Full-Wave Bridge Rectifier Circuit

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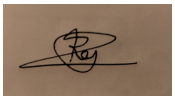
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Design and Performance Analysis of a Full-Wave Bridge Rectifier Circuit"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit Lab (07131102W)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

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Objective

- To design, construct, and verify a full-wave bridge rectifier circuit on a solderless breadboard using four silicon p-n junction diodes.
- To evaluate cycle-by-cycle conduction characteristics during both positive and negative half-cycles of an input AC signal.
- To observe, capture, and compare input and output voltage waveforms utilizing a Digital Storage Oscilloscope (DSO).
- To compute essential operating metrics including average DC output voltage (V_{DC}), RMS output voltage (V_{rms}), ripple factor (r), and rectification efficiency (η).
- To examine output voltage waveform smoothing using a parallel capacitive filter and compare empirical observations against theoretical predictions.

Introduction

Electronic devices demand a steady direct current (DC) power source for consistent operations, whereas commercial power distribution networks deliver alternating current (AC). Rectifier circuits convert bidirectional alternating signals into unidirectional pulsating outputs. Among fundamental rectifier configurations, the full-wave bridge topology is widely preferred due to its ability to utilize both half-cycles of the AC input without requiring a center-tapped transformer. Constructing the bridge configuration with four diodes arranged in a closed loop ensures continuous current flow across the load resistor in a single direction regardless of input polarity. Since unfiltered rectified outputs retain significant voltage ripple, this experiment further investigates the smoothing capabilities of a parallel reservoir filter capacitor.

Theory

Cycle-by-Cycle Conduction Analysis

Consider an AC voltage source supplying the bridge rectifier, defined mathematically as:

$$v_s(t) = V_m \sin(\omega t),$$

where $V_m = V_{rms}\sqrt{2}$ denotes the peak secondary voltage and $\omega = 2\pi f$ represents the angular frequency.

- **Positive Half-Cycle ($0 < \omega t < \pi$):** The upper secondary terminal operates at a higher potential relative to the lower terminal. Diodes D_1 and D_4 become forward-biased and conduct, whereas D_2 and D_3 remain reverse-biased (acting as open switches). Current flows from the source through D_1, passes top-to-bottom across load R_L, and returns via D_4.
- **Negative Half-Cycle ($\pi < \omega t < 2\pi$):** As input polarity reverses, D_2 and D_3 transition to forward-bias conduction while D_1 and D_4 become reverse-biased. The current path routes through D_2, flows across load R_L in the identical direction, and completes the loop via D_3.

Because current passes through the load in the same direction across both half-cycles, the output frequency doubles relative to the input line frequency: $f_{\text{out}} = 2f_{\text{in}}$.

Mathematical Formulation of Rectifier Parameters

Since two diodes conduct concurrently in each half-cycle, their combined forward voltage drops ($V_D \approx 0.7\,\text{V}$ per diode) reduce the peak output voltage:

$$V_{mo} = V_m - 2V_D.$$

Standard mathematical formulations yield the DC (average) and RMS output voltage values:

$$V_{DC} = \frac{2V_{mo}}{\pi}, \qquad V_{rms} = \frac{V_{mo}}{\sqrt{2}}.$$

The ripple factor measures residual AC fluctuations riding on the DC component. For an unfiltered full-wave rectifier:

$$r = \sqrt{\left(\frac{V_{rms}}{V_{DC}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} \approx 0.482.$$

The conversion efficiency η and Peak Inverse Voltage (PIV) ratings are expressed as:

$$\eta = \frac{P_{DC}}{P_{AC}} \times 100\%, \qquad \eta_{\max} = \frac{8}{\pi^2} \approx 81.2\%, \qquad \text{PIV} = V_m.$$

Effect of a Capacitive Filter

Connecting a filter capacitor C in parallel with load R_L reduces output ripple voltage. The capacitor charges rapidly to V_{mo} during input peaks and discharges slowly through R_L during conduction off-times. Assuming a linear discharge path, peak-to-peak ripple voltage $V_{r(p\text{-}p)}$ and ripple factor r follow:

$$V_{r(p\text{-}p)} \approx \frac{I_{DC}}{2fC} = \frac{V_{DC}}{2fR_LC}, \qquad r \approx \frac{1}{4\sqrt{3}\,fR_LC}.$$

Increasing load resistance R_L or capacitance C minimizes output voltage fluctuations, improving overall power delivery stability.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
Function Generator	Sine wave, 50 Hz, measured peak output 6 V	1
Silicon Rectifier Diodes	1N4007	4
Load Resistor R_L	965 Ω	1
Electrolytic Filter Capacitor C	47 µF	1
Dual-Channel Digital Storage Oscilloscope (DSO)	With 10× probes	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

Circuit Diagram

Experimental Procedure

1. **Bridge Assembly:** Mount four 1N4007 diodes on the breadboard following the bridge orientation shown in Figure 1, verifying correct terminal polarity.
2. **Source Integration:** Connect function generator leads across nodes AC_1 and AC_2, configured to deliver a 50 Hz sinusoidal voltage waveform.
3. **Unfiltered Baseline Testing:** Attach load resistor $R_L = 965\,\Omega$ across output terminals. Display input and output waveforms on DSO Channels 1 and 2 to record peak levels V_m and V_{mo}. Derive dependent parameters (V_{DC}, V_{rms}, r, η) using analytical relationships.
4. **Filtered Analysis:** Connect electrolytic capacitor $C = 47\,\mu F$ across load R_L, observing waveform smoothing on the DSO. Determine filtered V_{DC} and ripple $V_{r(p\text{-}p)}$ values.

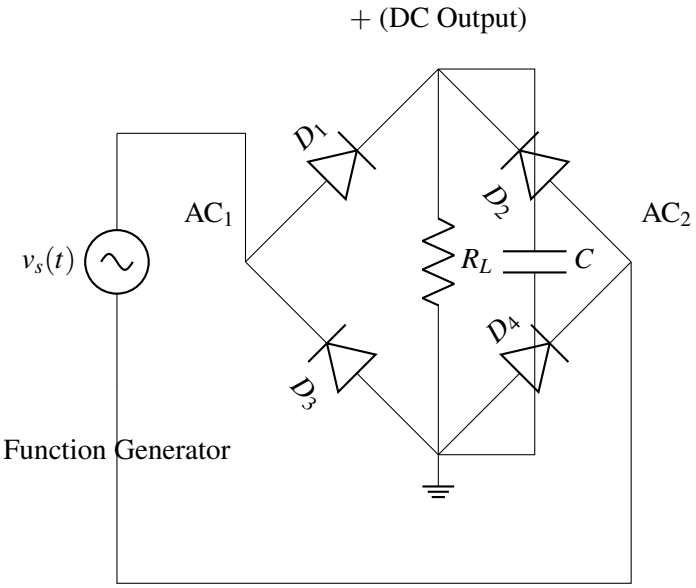


Figure 1: Full-wave bridge rectifier circuit driven by an AC source with parallel load R_L and optional capacitor filter C.

Theoretical Calculation

Using measured laboratory parameter values ($V_m = 6\,\text{V}$, $V_D = 0.7\,\text{V}$, $f = 50\,\text{Hz}$, and $R_L = 965\,\Omega$):

$$V_{mo} = V_m - 2V_D = 6 - 1.4 = 4.60\,\text{V},$$

$$V_{DC} = \frac{2V_{mo}}{\pi} = \frac{2 \times 4.60}{3.1416} = 2.928\,\text{V}, \qquad I_{DC} = \frac{V_{DC}}{R_L} = \frac{2.928}{965} = 3.03\,\text{mA},$$

$$V_{rms(\text{out})} = \frac{V_{mo}}{\sqrt{2}} = \frac{4.60}{1.4142} = 3.253\,\text{V},$$

$$\eta = \frac{V_{DC}^2/R_L}{V_{rms(\text{out})}^2/R_L} \times 100\% = \frac{2.928^2}{3.253^2} \times 100\% \approx 81.0\%, \qquad \text{PIV} = V_m = 6\,\text{V}.$$

Introducing the 47 µF filter capacitor yields expected peak-to-peak ripple voltage:

$$V_{r(p\text{-}p)} \approx \frac{V_{DC}}{2fR_LC} = \frac{2.928}{2 \times 50 \times 965 \times 47 \times 10^{-6}} \approx 0.646\,\text{V}.$$

Data Collection

Table 2: Unfiltered Rectifier Performance

Load R_L	Peak Input V_m	Peak Output V_{mo}	Measured V_{DC}	Measured V_{rms}	Ripple Factor r	Efficiency η
965 Ω	6 V	2 V	1.27 V	1.41 V	0.483	81.1%

Table 3: Filtered Rectifier Performance: Effect of the 47 µF Capacitor

Load R_L	Filter C	Measured V_{DC}	Ripple $V_{r(p\text{-}p)}$
965 Ω	47 µF	1.78 V	0.44 V

Oscilloscope Waveforms

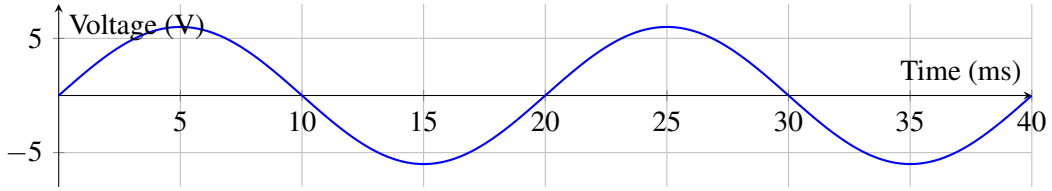


Figure 2: Sinusoidal AC input voltage trace ($V_m = 6\,\text{V}$, $f = 50\,\text{Hz}$).

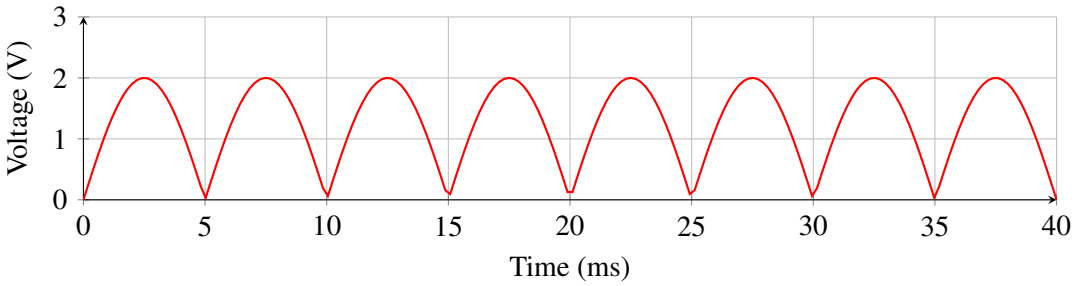


Figure 3: Unfiltered full-wave rectified output voltage trace ($V_{mo} = 2\,\text{V}$, $f = 100\,\text{Hz}$).

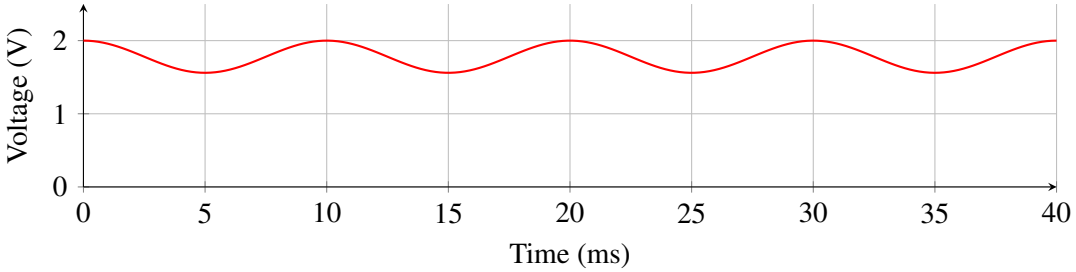


Figure 4: Filtered DC output voltage trace demonstrating smoothed output waveform.

Result Analysis

Experimental measurement yielded a peak output voltage of $V_{mo} = 2\,\text{V}$, exhibiting a variance relative to theoretical calculations (4.60 V). Dimensionless shape parameters aligned closely with theoretical predictions: the observed ripple factor (0.483) matched theoretical limits (0.482), while efficiency reached 81.1% against an ideal maximum of 81.2%. Adding the 47 µF filter capacitor increased V_{DC} from 1.27 V to 1.78 V while attenuating peak-to-peak ripple to 0.44 V, confirming successful waveform filtering.

Conclusion

The full-wave bridge rectifier circuit was constructed and evaluated under AC excitation ($V_m = 6\,\text{V}$, $f = 50\,\text{Hz}$). Full-wave rectification was successfully verified through comparative waveform measurements and quantitative parameter evaluation. Adding a parallel filter capacitor effectively smoothed voltage variations, elevating average DC output voltage while reducing ripple components.

References

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